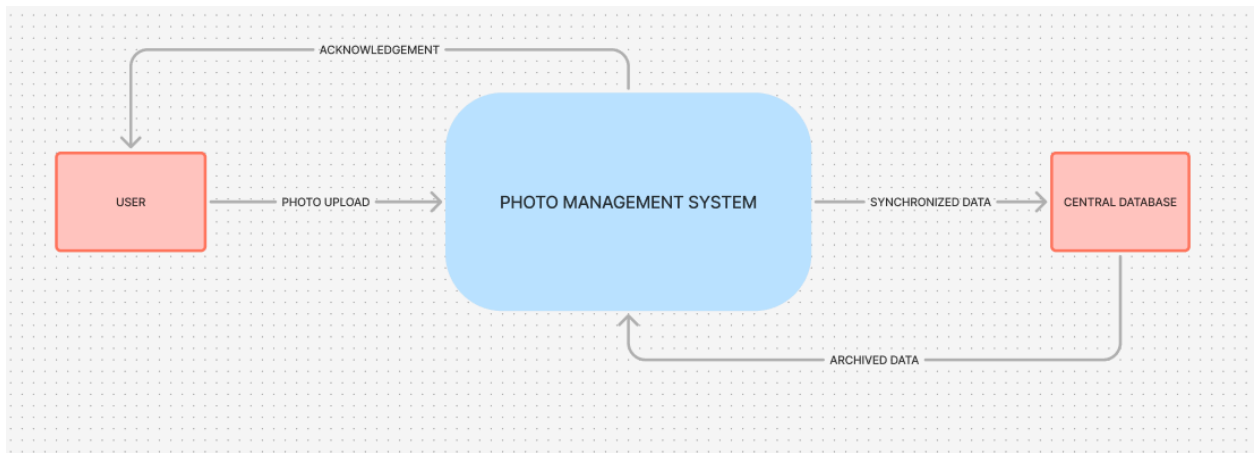


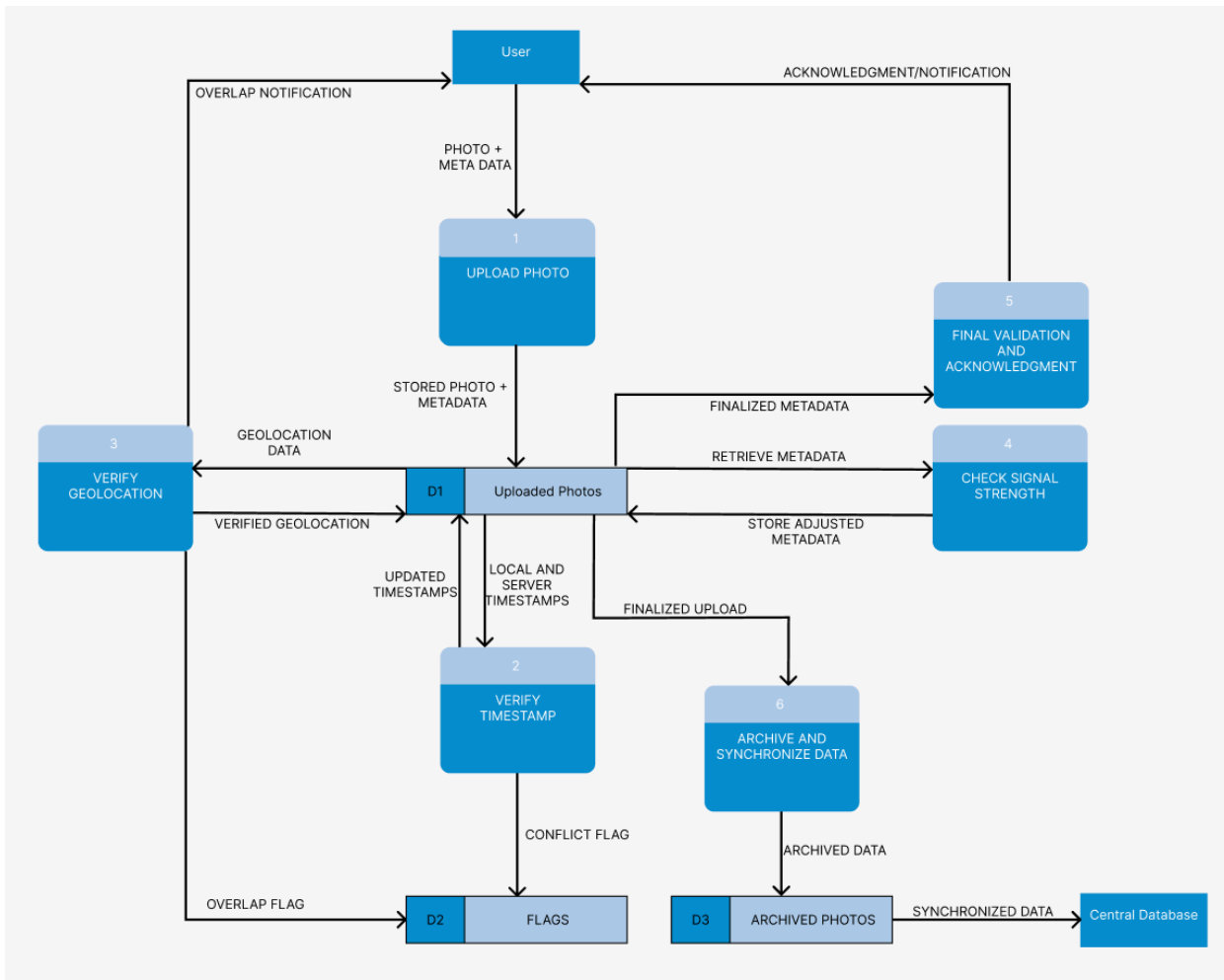
Design Specifications

Context Diagram

Level 0 Data Flow Diagram (DFD):



Level 1 Flow Diagram (DFD):



- **Level 0:** Users report waste; data is sent to a server, processed, and shared with LGUs.
- **Level 1:** Includes details of user authentication, GPS data processing, and reward system integration.

Step 1: User uploads a photo

- **Process:** Upload Photo (Rounded Rectangle)
 - Accepts the uploaded photo and stores it with metadata.
- **Data Store:** Uploaded Photos (Open Rectangle)
 - Stores the photo and its metadata for further processing.
- **Entity:** User (Square)
 - External person uploading the photo.
- **Data Flow:**
 - From **User** to **Upload Photo**: "Photo + Metadata."
 - From **Upload Photo** to **Uploaded Photos**: "Stored Photo + Metadata."

Step 2: System verifies the timestamp

- **Process:** Verify Timestamp (Rounded Rectangle)
 - Compares local and server timestamps and flags conflicts.
- **Data Store:** Uploaded Photos (Open Rectangle)
 - Supplies local and server timestamps to the process.
 - Stores updated timestamps after verification.
- **Data Store:** Flags (Open Rectangle)
 - Stores flagged cases where conflicts exist.
- **Data Flow:**
 - From **Uploaded Photos** to **Verify Timestamp**: "Local and Server Timestamps."
 - From **Verify Timestamp** to **Uploaded Photos**: "Updated Timestamps."
 - From **Verify Timestamp** to **Flags**: "Conflict Flag."

Step 3: System verifies geolocation

- **Process:** Verify Geolocation (Rounded Rectangle)
 - Checks for overlapping geolocation data and flags issues.
- **Data Store:** Uploaded Photos (Open Rectangle)
 - Supplies geolocation data for verification.
 - Stores validated geolocation after processing.
- **Data Store:** Flags (Open Rectangle)
 - Stores flagged cases of geolocation overlap.
- **Entity:** User (Square)
 - Notified if their upload conflicts with another.
- **Data Flow:**
 - From **Uploaded Photos** to **Verify Geolocation**: "Geolocation Data."
 - From **Verify Geolocation** to **Uploaded Photos**: "Verified Geolocation."
 - From **Verify Geolocation** to **Flags**: "Overlap Flag."
 - From **Verify Geolocation** to **User**: "Overlap Notification."

Step 4: System checks signal strength

- **Process:** Signal Strength Check (Rounded Rectangle)
 - Evaluates metadata to detect weak signal scenarios and adjusts timestamps if necessary.
- **Data Store:** Uploaded Photos (Open Rectangle)
 - Supplies metadata (timestamp and other relevant data) for signal strength evaluation.
 - Receives adjusted metadata after processing.
- **Data Flow:**
 - From **Uploaded Photos** to **Signal Strength Check**: "Retrieve Metadata."
 - From **Signal Strength Check** to **Uploaded Photos**: "Store Adjusted Metadata."

Step 5: Final Validation and Acknowledgment

- **Process:** Validate and Acknowledge Upload (Rounded Rectangle)

- Performs a final review of the uploaded photo's metadata (timestamp, geolocation, and signal strength) to ensure all issues are resolved.
- Sends acknowledgement or notification to the user.
- **Data Store:** Uploaded Photos (Open Rectangle)
 - Supplies metadata for final validation.
- **Entity:** User (Square)
 - Receives acknowledgment or notification of the final status of their upload.
- **Data Flow:**
 - From **Uploaded Photos** to **Validate and Acknowledge Upload**: "Finalized Metadata."
 - From **Validate and Acknowledge Upload** to **User**: "Acknowledgment/Notification."

Step 6: Archive and Synchronize Data

- **Process:** Archive and Synchronize Data (Rounded Rectangle)
 - Archives validated photo uploads into permanent storage.
 - Synchronizes the archived data with external or central systems (e.g., a cloud database or central repository).
- **Data Store:** Archived Photos (Open Rectangle)
 - Stores finalized and validated photo uploads for long-term use or reference.
- **Entity:** Central Database (Square)
 - Receives synchronized data for central processing, analytics, or storage.
- **Data Flow:**
 - From **Uploaded Photos** to **Archive and Synchronize Data**: "Finalized Upload."
 - From **Archive and Synchronize Data** to **Archived Photos**: "Archived Data."
 - From **Archive and Synchronize Data** to **Central Database**: "Synchronized Data."

Data Dictionary

DATA NAME	DESCRIPTION	SOURCE	DESTINATION	FORMAT/ ATTRIBUTES
Photo + Metadata	The photo and its metadata are saved for processing.	Upload Photo	Uploaded Photos	Photo file, timestamp, geolocation, signal strength
Stored Photo + Metadata	Timestamp information used to verify conflicts.	Uploaded Photos	Verify Timestamp	Local timestamp, server timestamp
Local and Server Timestamps	Revised timestamp data after verification.	Verify Timestamp	Uploaded Photos	Adjusted local and server timestamps
Updated Timestamps	Flags created for timestamp conflicts.	Verify Timestamp	Flags	Conflict ID, timestamp conflict details
Conflict Flag	Metadata related to the location where the photo	Uploaded Photos	Verify Geolocation	Latitude, longitude, altitude

	was taken.			
Geolocation Data	Validated geolocation metadata after processing.	Verify Geolocation	Uploaded Photos	Latitude, longitude, altitude
Verified Geolocation	Validated geolocation metadata after processing.	Verify Geolocation	Uploaded Photos	Latitude, longitude, altitude
Overlap Flag	Flags created for overlapping geolocation data conflicts.	Verify Geolocation	Flags	Conflict ID, geolocation overlap details
Overlap Notification	Notification sent to the user about geolocation overlap.	Verify Geolocation	User	Message detailing geolocation overlap
Retrieve Metadata	Metadata retrieved for signal strength evaluation.	Uploaded Photos	Signal Strength Check	Timestamp, geolocation, signal strength
Store Adjusted Metadata	Adjusted metadata after evaluating signal strength.	Signal Strength Check	Uploaded Photos	Adjusted timestamp, geolocation, or signal strength
Finalized Metadata	Fully validated metadata, ready for final acknowledgment.	Uploaded Photos	Validate and Acknowledge Upload	Finalized timestamp, geolocation, and signal strength
Acknowledgment/Notification	Acknowledgment or notification sent to the user after the final validation.	Validate and Acknowledge Upload	User	Message confirming upload status (success or failure)
Finalized Upload	Fully validated and finalized photo with metadata ready for archival.	Uploaded Photos	Archive and Synchronize Data	Photo file, finalized metadata
Archived Data	Photo and metadata stored for long-term use.	Archive and Synchronize Data	Archived Photos	Photo file, finalized metadata
Synchronized Data	Finalized data sent to a central repository for storage or analytics.	Archive and Synchronize Data	Central Database	Photo file, metadata, and synchronization confirmation